Design and Analysis of Randomized Trials in Public Health

Homework 6

Please use the "hw4" SAS dataset on Brightspace to answer **questions 1-6** below. Before importing the dataset into SAS, run the code below. This will create value labels for your dataset.

```
proc format;
      value yes12no       1="yes"
                          2="no";
      value sex           1="female"
                          2="male";
      value trt           1="treatment"
                          0="control";
run;
```

These data are from a randomized trial in which eligible participants were randomized to receive a physical activity intervention ("treatment") or no intervention ("control"). Their body mass index (BMI) was measured at the beginning ("baselineBMI") and end ("endlineBMI") of the study. Participants were eligible for the study if they (1) were aged 18-50 years, (2) were evaluated to have a sedentary lifestyle, and (3) provided informed consent. Participants were excluded if they failed to meet any of these criteria.

1. Does the physical activity intervention affect body mass index? Use "endlineBMI" as your outcome, and answer this question using a nonparametric test. State which test you used, what your null hypothesis was, what conclusion you reached (in lay, not statistical, terms), and how you reached that conclusion (test statistic/p-value). (4 points)

2. People are considered overweight if their BMI is 25 or greater. Create an indicator variable that identifies whether a person is overweight or not at the end of the study. When you do this, make sure you correctly handle missing values. Does the physical activity intervention change a person's chances of being overweight? Use a chi-square test of independence to answer this question. Clarify what your null hypothesis was, what conclusion you reached (in lay, not statistical, terms), and how you reached that conclusion (test statistic/p-value). (3 points)

3. Answer #2 using a nonparametric test. Clarify which test you used, what your null hypothesis was, what conclusion you reached (in lay, not statistical, terms), and how you reached that conclusion (test statistic/p-value). (4 points)

4. Does the BMI of people who receive no intervention change over time? Use a parametric test to answer this question. Clarify which test you used, what your null hypothesis was, what conclusion you reached (in lay, not statistical, terms), and how you reached that conclusion (test statistic/p-value). (4 points)

5. Does the BMI of people who receive the physical activity intervention change over time? Use a nonparametric test to answer this question. Clarify which test you used, what your null hypothesis was, what conclusion you reached (in lay, not statistical, terms), and how you reached that conclusion (test statistic/p-value). (4 points)

6. You want to evaluate a new integrated diet and physical activity intervention in a similar population using a randomized trial with a completely randomized design. This intervention will be compared to a no-intervention control. You want to detect a difference in BMI of at least 1 kg/m^2 between the two treatments. What sample size would you need to have at least 80% power? 90% power? Use the BMI data at baseline to help you conduct the power calculations. Assume a two-sided hypothesis and a significance level of 0.05. (3 points)

7. [Note this question is not related to the others above.] You have designed an intervention that aims to reduce premature birth in a vulnerable population. You will conduct a randomized trial in which you will compare pregnancy outcomes of women receiving your intervention to those of women receiving standard antenatal care. Your outcome of interest is gestational age at delivery, measured in weeks. You conduct a power calculation assuming a 0.05 significance level and determine you need 253 women in each study arm.

a) If you increase the number of women in each study arm, will your statistical power increase or decrease? (1 point)

b) You decide you want to detect a smaller effect size than you used in your original power calculations. You will still use the same sample size. Will your statistical power now increase or decrease? (1 point)

c) You decide that you would rather use a 0.01 significance level. Will your statistical power now increase or decrease? (1 point)

Homework 6

Alexandra Chang
PUBH 526: Design and Analysis of Randomized Trials in Public Health

October 17, 2025

Please use the "hw4" SAS dataset on Brightspace to answer the questions below. Before importing the dataset into SAS, run the code below. This will create value labels for your dataset.

```
proc format;
   value yes12no        1="yes"
                        2="no";
   value sex            1="female"
                        2="male";
   value trt            1="treatment"
                        0="control";
run;
```

These data are from a randomized trial in which eligible participants were randomized to receive a physical activity intervention ("treatment") or no intervention ("control"). Their body mass index (BMI) was measured at the beginning ("baselineBMI") and end ("endlineBMI") of the study. Participants were eligible for the study if they (1) were aged 18-50 years, (2) were evaluated to have a sedentary lifestyle, and (3) provided informed consent. Participants were excluded if they failed to meet any of these criteria.

Question 1 (4 points)

Does the physical activity intervention affect body mass index? Use "endlineBMI" as your outcome, and answer this question using a nonparametric test. State which test you used, what your null hypothesis was, what conclusion you reached (in lay, not statistical, terms), and how you reached that conclusion (test statistic/p-value).

Test Statistic	Value	DF	p-value
Z (Wilcoxon Two–Sample)	2.6504	–	0.0080
Kruskal–Wallis Chi–Square	7.0267	1	0.0080

- H_0: The distributions of endline BMI for the treatment and control groups are equal.
- H_A: The distributions differ (the intervention affects BMI).
- **Decision rule:** Reject H_0 if $p < 0.05$.
- **Decision:** $p = 0.008 < 0.05 \Rightarrow$ **Reject** H_0.
  - In the Wilcoxon output, the **two-sided** p-value ($p = 0.008$) was used because the hypothesis tested whether the intervention **affected** BMI in either direction, not specifically increased or decreased it.

Conclusion: Participants who received the physical activity intervention had significantly different BMI values at the end of the study compared with those in the control group. This indicates that the intervention meaningfully affected participants' body mass index.

Question 2 (3 points)

People are considered overweight if their BMI is 25 or greater. Create an indicator variable that identifies whether a person is overweight or not at the end of the study. When you do this, make sure you correctly handle missing values. Does the physical activity intervention change a person's chances of being overweight? Use a chi-square test of independence to answer this question. Clarify what your null hypothesis was, what conclusion you reached (in lay, not statistical, terms), and how you reached that conclusion (test statistic/p-value).

Statistic	Value	DF	p-value
Chi–Square	2.7389	1	0.0979
Likelihood–Ratio Chi–Square	2.7602	1	0.0966
Continuity–Adjusted Chi–Square	2.3694	1	0.1237
Fisher's Exact (two–sided)	–	–	0.1062

- H_0: Overweight status is independent of treatment group.
- H_A Overweight status differs between treatment and control groups.
- **Decision rule:** Reject H_0 if $p < 0.05$.
- **Decision:** $p = 0.0979 > 0.05 \Rightarrow$ **Fail to reject** H_0.

Conclusion: The difference in the proportion of participants who were overweight at the end of the study between the treatment and control groups was **not** statistically significant. This suggests that the physical–activity intervention did **not** have a significant effect on the likelihood of being overweight.

Question 3 (3 points)

Answer 2 using a <u>nonparametric</u> test. Clarify which test you used, what your null hypothesis was, what conclusion you reached (in lay, not statistical, terms), and how you reached that conclusion (test statistic/p-value).

Statistic	p-value
Fisher's Exact (two–sided)	0.1062

- H_0: Overweight status and treatment group are independent.
- H_A: Overweight status differs between treatment and control groups.
- **Decision rule:** Reject H_0 if $p < 0.05$.
- **Decision:** $p = 0.1062 > 0.05 \Rightarrow$ **Fail to reject** H_0.

Conclusion: The Fisher's Exact Test found **no** significant difference in the proportion of overweight participants between the treatment and control groups. This suggests that the physical–activity intervention did **not** meaningfully affect the likelihood of being overweight.

Question 4 (4 points)

Does the BMI of people <u>who receive no intervention</u> change over time? Use a parametric test to answer this question. Clarify which test you used, what your null hypothesis was, what conclusion you reached (in lay, not statistical, terms), and how you reached that conclusion (test statistic/p-value).

Statistic	Mean Difference	DF	t Value	p-value
Paired t–test	-0.3661	185	-6.25	<0.0001

- H_0: The mean BMI at baseline equals the mean BMI at endline (no change over time).
- H_A: The mean BMI at baseline and endline differ (BMI changed over time).
- **Decision rule:** Reject H_0 if $p < 0.05$.
- **Decision:** $t = -6.25$, $p < 0.0001 < 0.05 \Rightarrow$ **Reject** H_0.

Conclusion: Among participants who received no intervention, BMI significantly decreased from baseline to endline. This suggests that even without the physical–activity program, participants' BMI showed a small but statistically significant reduction over time.

Question 5 (4 points)

Does the BMI of people who receive the physical activity intervention change over time? Use a nonparametric test to answer this question. Clarify which test you used, what your null hypothesis was, what conclusion you reached (in lay, not statistical, terms), and how you reached that conclusion (test statistic/p-value).

Statistic	N	Signed Rank (S)	p-value
Wilcoxon Signed–Rank	217	-11346.5	<0.0001

- H_0**:** The median difference in BMI (endline – baseline) equals 0 (no change over time).
- H_A**:** The median difference in BMI differs from 0 (BMI changed over time).
- **Decision rule:** Reject H_0 if $p < 0.05$.
- **Decision:** $S = -11346.5$, $p < 0.0001 < 0.05 \Rightarrow$ **Reject** H_0.

Conclusion: Among participants who received the physical–activity intervention, BMI significantly decreased from baseline to endline. This indicates that the intervention led to a meaningful reduction in participants' body mass index over time.

Question 6 (3 points)

You want to evaluate a new integrated diet and physical activity intervention in a similar population using a randomized trial with a completely randomized design. This intervention will be compared to a no-intervention control. You want to detect a difference in BMI of at least 1 kg/m2 between the two treatments. What sample size would you need to have at least 80% power? 90% power? Use the BMI data at baseline to help you conduct the power calculations. Assume a two-sided hypothesis and a significance level of 0.05.

Target power	Effect size $d = \frac{\Delta}{\hat{\sigma}}$	Total N	N per arm
80%	0.5578	104	52
90%	0.5578	138	69

- H_0**:** Mean BMI is equal between intervention and control groups.
- H_A**:** Mean BMI differs between groups by at least 1 kg/m^2.
- **Decision rule:** Target 80% or 90% power at $\alpha = 0.05$ to detect $\Delta = 1$.

Conclusion: Based on the estimated baseline BMI standard deviation ($\hat{\sigma} = 1.7933$), a total of 104 participants (52 per arm) would provide 80% power, and 138 participants (69 per arm) would provide 90% power to detect a 1 kg/m^2 difference. If anticipating 10% attrition, the required sample size per arm should be inflated to approximately 58 and 77, respectively.

[Note this question is not related to the others above.] You have designed an intervention that aims to reduce premature birth in a vulnerable population. You will conduct a randomized trial in which you will compare pregnancy outcomes of women receiving your intervention to those of women receiving standard antenatal care. Your outcome of interest is gestational age at delivery, measured in weeks. You conduct a power calculation assuming a 0.05 significance level and determine you need 253 women in each study arm.

Question 7.a (1 point)

If you increase the number of women in each study arm, will your statistical power increase or decrease?

- Statistical power will **increase**.
- Increasing the sample size reduces the standard error, making it easier to detect a true effect if it exists. With larger groups, even small differences become more statistically detectable.

Question 7.b (1 point)

You decide you want to detect a smaller effect size than you used in your original power calculations. You will still use the same sample size. Will your statistical power now increase or decrease?

- Statistical power will **decrease**.
- Smaller effect sizes are harder to detect with the same level of variability. If the sample size stays constant but the target effect becomes smaller, the likelihood of detecting it (i.e., power) decreases.

Question 7.c (1 point)

You decide that you would rather use a 0.01 significance level. Will your statistical power now increase or decrease?

- Statistical power will **decrease**.
- Lowering the significance level (from 0.05 to 0.01) makes the test more conservative. It becomes harder to reject the null hypothesis, so the probability of detecting a true effect (i.e., power) decreases.

See SAS codes/outputs below:

```
/*************************************************************************

 * PUBH 526     Design and Analysis of Randomized Trials in Public Health
 * Homework 6
 * Alexandra Chang
 * Dataset: hw4.sas7bdat
 *************************************************************************/


libname mydata '/home/u64258127/sasuser.v94';

proc format;
    value yes12no  1 = "yes"  2 = "no";
    value sex      1 = "female"  2 = "male";
    value trt      1 = "treatment"  0 = "control";
run;


/*************************************************************************

 * Question 1: Does the physical activity intervention affect BMI?
 * Outcome: endlineBMI
 * Comparison: treatment vs. control
 * Test: Nonparametric (Wilcoxon Rank-Sum Test)
 *************************************************************************/


proc npar1way data=mydata.hw4 wilcoxon;
    class treatment;
    var endlineBMI;
    format treatment trt.;
    title "Question 1: Wilcoxon Rank-Sum Test for Endline BMI by Treatment Group";
run;


/*************************************************************************

 * Question 2: Does the intervention change a persons chance of being overweight?
 * Outcome: Overweight status at endline (BMI     25)
 * Test: Chi-Square Test of Independence
 *************************************************************************/


```

```
/* Create indicator variable for overweight status */
data hw4_overwt;
    set mydata.hw4;
    if not missing(endlineBMI) then do;
        if endlineBMI >= 25 then overweight = 1;
        else overweight = 0;
    end;
run;

/* Frequency table and Chi-Square test */
proc freq data=hw4_overwt;
    tables treatment*overweight / chisq expected norow nocol nopercent;
    format treatment trt.;
    title "Question 2: Chi-Square Test of Independence for Overweight Status by Treatment Group";
run;


/**************************************************************************
 * Question 3: Nonparametric test for overweight status by treatment group
 * Outcome: Overweight status at endline (BMI     25)
 * Test: F i s h e r s Exact Test (nonparametric alternative to Chi-Square)
 **************************************************************************/

proc freq data=hw4_overwt;
    tables treatment*overweight / fisher expected norow nocol nopercent;
    format treatment trt.;
    title "Question 3: Fisher's Exact Test for Overweight Status by Treatment Group";
run;


/**************************************************************************
 * Question 4: Does BMI change over time among the control group?
 * Dataset: hw4.sas7bdat
 * Test: Paired t-test (parametric)
 **************************************************************************/

/* Restrict to control group only */
```

```
proc ttest data=mydata.hw4;
    where treatment = 0;
    paired endlineBMI*baselineBMI;
    format treatment trt.;
    title "Question 4: Paired t-test for Change in BMI Over Time
        Control Group Only";
run;


/**********************************************************************
 * Question 5: Does BMI change over time among the treatment group?
 * Dataset: hw4.sas7bdat
 * Test: Wilcoxon Signed-Rank Test (nonparametric paired test)
 **********************************************************************/

/* Step 1: Create the difference variable */
data trt_only;
    set mydata.hw4;
    where treatment = 1;
    diff_bmi = endlineBMI - baselineBMI;
run;

/* Step 2: Perform Wilcoxon Signed-Rank Test */
proc univariate data=trt_only mu0=0;
    var diff_bmi;
    title "Question 5: Wilcoxon Signed-Rank Test for Change in BMI
           Treatment Group Only";
run;


/**********************************************************************
 * Question 6: Sample size for a completely randomized 2-arm RCT
 * Goal: Detect    = 1 kg/m^2 difference in BMI (two-sided    =
    0.05)
 * Use baseline BMI to estimate SD, assume equal SD in both arms
 **********************************************************************/

/**********************************************************************
 * Step 1: Estimate SD of baseline BMI (overall)
 **********************************************************************/
proc means data=mydata.hw4 n mean stddev;
```

```
    var baselineBMI;
    title "Question 6: Baseline BMI Summary (Use SD for Power Calculation)";
run;

/***************************************************************************
 * Step 2: Power analysis for 2-sample comparison of means
 * Goal: Detect     = 1 kg/m^2 difference in BMI
 * Assumptions:
 *   - Equal allocation (1:1)
 *   -     = 0.05 (two-sided)
 *   -     = 1.7933 (from Step 1)
 ***************************************************************************/

proc power;
    twosamplemeans
        test = diff
        meandiff = 1           /*     = 1 kg/m^2 */
        stddev   = 1.7933      /* estimated from baseline BMI */
        alpha    = 0.05
        power    = 0.80 0.90   /* solve for these power levels */
        sides    = 2
        ntotal   = .;          /* tell SAS to compute total sample size */
    title "Question 6: Required Sample Size for     = 1 in BMI (Two-Sided     = 0.05)";
run;

/* Compute N per arm */
data Q6_final;
    set Q6_power;
    N_per_arm = ceil(NTotal/2);
run;

proc print data=Q6_final noobs label;
    label N_per_arm = "Required N per arm (rounded up)";
    title "Question 6: Final Sample Size per Arm";
run;
```

Listing 1: SAS Code for Homework 6

Question 1: Wilcoxon Rank-Sum Test for Endline BMI by Treatment Group

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable endlineBMI Classified by Variable treatment					
treatment	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
treatment	217	40744.50	43834.0	1165.50336	187.762673
control	186	40661.50	37572.0	1165.50336	218.610215
Average scores were used for ties.					

Wilcoxon Two-Sample Test					
				t Approximation	
Statistic	Z	Pr > Z	Pr > |Z|	Pr > Z	Pr > |Z|
40661.50	2.6504	0.0040	0.0080	0.0042	0.0084
Z includes a continuity correction of 0.5.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
7.0267	1	0.0080

Question 2: Chi-Square Test of Independence for Overweight Status by Treatment Group

The FREQ Procedure

Frequency Expected

Table of treatment by overweight			
	overweight		
treatment(Treatment)	0	1	Total
control	39 46.154	147 139.85	186
treatment	61 53.846	156 163.15	217
Total	100	303	403
Frequency Missing = 97			

Statistics for Table of treatment by overweight

Statistic	DF	Value	Prob
Chi-Square	1	2.7389	0.0979
Likelihood Ratio Chi-Square	1	2.7602	0.0966
Continuity Adj. Chi-Square	1	2.3694	0.1237
Mantel-Haenszel Chi-Square	1	2.7321	0.0983
Phi Coefficient		-0.0824	
Contingency Coefficient		0.0822	
Cramer's V		-0.0824	

Fisher's Exact Test	
Cell (1,1) Frequency (F)	39
Left-sided Pr <= F	0.0615
Right-sided Pr >= F	0.9621
Table Probability (P)	0.0236
Two-sided Pr <= P	0.1062

Sample Size = 403
Frequency Missing = 97

WARNING: 19% of the data are missing.

Question 3: Fisher's Exact Test for Overweight Status by Treatment Group

The FREQ Procedure

Frequency Expected

Table of treatment by overweight			
	overweight		
treatment(Treatment)	0	1	Total
control	39 46.154	147 139.85	186
treatment	61 53.846	156 163.15	217
Total	100	303	403
Frequency Missing = 97			

Statistics for Table of treatment by overweight

Statistic	DF	Value	Prob
Chi-Square	1	2.7389	0.0979
Likelihood Ratio Chi-Square	1	2.7602	0.0966
Continuity Adj. Chi-Square	1	2.3694	0.1237
Mantel-Haenszel Chi-Square	1	2.7321	0.0983
Phi Coefficient		-0.0824	
Contingency Coefficient		0.0822	
Cramer's V		-0.0824	

Fisher's Exact Test	
Cell (1,1) Frequency (F)	39
Left-sided Pr <= F	0.0615
Right-sided Pr >= F	0.9621
Table Probability (P)	0.0236
Two-sided Pr <= P	0.1062

Sample Size = 403
Frequency Missing = 97

WARNING: 19% of the data are missing.

Question 4: Paired t-test for Change in BMI Over Time – Control Group Only

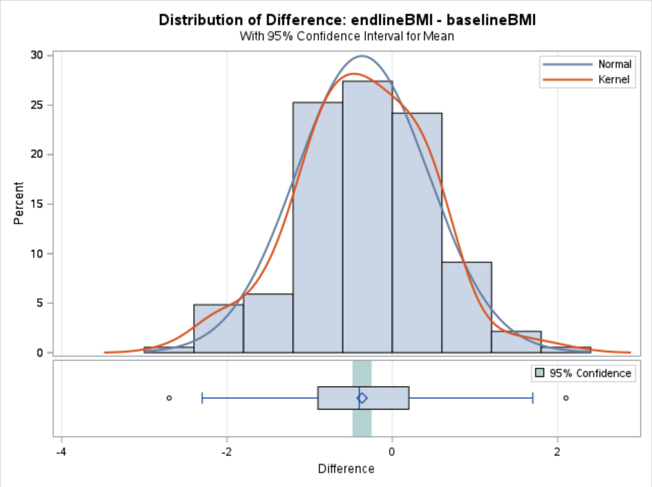
The TTEST Procedure

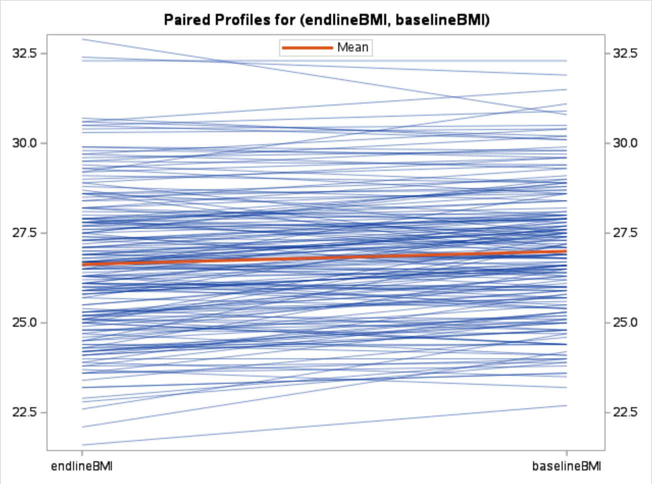
Difference: endlineBMI - baselineBMI

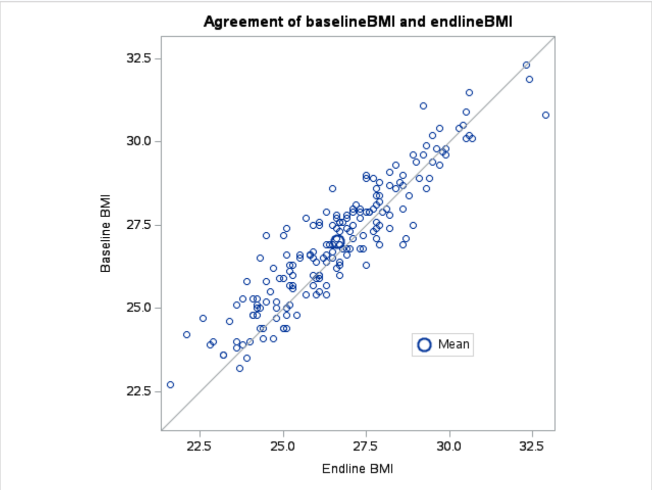
N	Mean	Std Dev	Std Err	Minimum	Maximum
186	-0.3661	0.7987	0.0586	-2.7000	2.1000

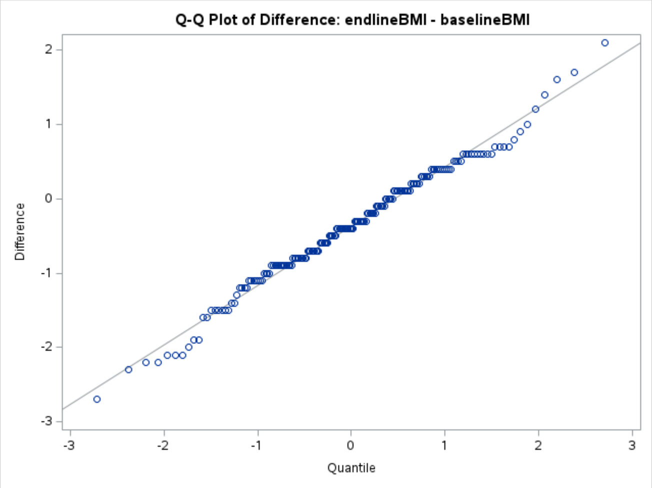
Mean	95% CL Mean		Std Dev	95% CL Std Dev	
-0.3661	-0.4817	-0.2506	0.7987	0.7249	0.8893

DF	t Value	Pr > |t|
185	-6.25	<.0001









Question 5: Wilcoxon Signed-Rank Test for Change in BMI – Treatment Group Only

The UNIVARIATE Procedure
Variable: diff_bmi

Moments			
N	217	Sum Weights	217
Mean	-1.2009217	Sum Observations	-260.6
Std Deviation	0.72072435	Variance	0.51944359
Skewness	0.2461841	Kurtosis	0.01218208
Uncorrected SS	425.16	Corrected SS	112.199816
Coeff Variation	-60.014269	Std Error Mean	0.04892596

Basic Statistical Measures			
Location		Variability	
Mean	-1.20092	Std Deviation	0.72072
Median	-1.20000	Variance	0.51944
Mode	-1.30000	Range	4.00000
		Interquartile Range	1.00000

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	-24.5457	Pr > |t|	<.0001
Sign	M	-96.5	Pr >= |M|	<.0001
Signed Rank	S	-11346.5	Pr >= |S|	<.0001

Quantiles (Definition 5)	
Level	Quantile
100% Max	1.0
99%	0.5
95%	0.1
90%	-0.2
75% Q3	-0.7
50% Median	-1.2
25% Q1	-1.7
10%	-2.1
5%	-2.3
1%	-2.7
0% Min	-3.0

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
-3.0	95	0.3	193
-2.9	128	0.5	76
-2.7	109	0.5	103
-2.7	98	0.9	48
-2.6	114	1.0	160

Question 6: Baseline BMI Summary (Use SD for Power Calculation)

The MEANS Procedure

Analysis Variable : baselineBMI Baseline BMI		
N	Mean	Std Dev
403	27.1191067	1.7933285

Question 6: Required Sample Size for Δ = 1 in BMI (Two-Sided α = 0.05)

The POWER Procedure
Two-Sample t Test for Mean Difference

Fixed Scenario Elements	
Distribution	Normal
Method	Exact
Number of Sides	2
Alpha	0.05
Mean Difference	1
Standard Deviation	1.7933
Null Difference	0
Group 1 Weight	1
Group 2 Weight	1

Computed N Total			
Index	Nominal Power	Actual Power	N Total
1	0.8	0.804	104
2	0.9	0.902	138

Question 6: Final Sample Size per Arm

Analysis	Index	Sides	Alpha	Mean Diff	Std Dev	Weight1	Weight2	Nominal Power	Null Diff	Actual Power	N Total	Error	Information	Required N per arm (rounded up)
TwoSampleMeans	1	2	0.05	1	1.79	1	1	0.8	0	0.804	104			52
TwoSampleMeans	2	2	0.05	1	1.79	1	1	0.9	0	0.902	138			69

Design and Analysis of Randomized Trials in Public Health

Homework 6 Solutions

Please use the "hw4" SAS dataset on Brightspace to answer **questions 1-6** below. Before importing the dataset into SAS, run the code below. This will create value labels for your dataset. There is an accompanying video that also gives you some tips on completing this homework.

```
proc format;
      value yes12no      1="yes"
                         2="no";
      value sex          1="female"
                         2="male";
      value trt          1="treatment"
                         0="control";
run;
```

These data are from a randomized trial in which eligible participants were randomized to receive a physical activity intervention ("treatment") or no intervention ("control"). Their body mass index (BMI) was measured at the beginning ("baselineBMI") and end ("endlineBMI") of the study. Participants were eligible for the study if they (1) were aged 18-50 years, (2) were evaluated to have a sedentary lifestyle, and (3) provided informed consent. Participants were excluded if they failed to meet any of these criteria.

1. Does the physical activity intervention affect body mass index? Use "endlineBMI" as your outcome, and answer this question using a nonparametric test. State which test you used, what your null hypothesis was, what conclusion you reached (in lay, not statistical, terms), and how you reached that conclusion (test statistic/p-value). (4 points)

Use a Wilcoxon rank-sum test (the normal approximation or t approximation are both fine and not very different, since we have a large sample size; use the two-sided p-values; Kruskal-Wallis is also fine to do – relevant p-values are highlighted below)

H_0: the distribution of BMI is the same whether or not people in this population receive the physical activity intervention

p=0.0080 (for the two-sided t), which is less than α=0.05

This study provides evidence at the 0.05 significance level that the distribution of BMI in this population differs depending on whether or not people receive the physical activity intervention.

```
proc npar1way data=hw4 wilcoxon;
      class treatment;
      var endlineBMI;
run;
```

Wilcoxon Two-Sample Test					
Statistic	Z	Pr > Z	Pr > |Z|	t Approximation	
				Pr > Z	Pr > |Z|
40661.50	2.6504	0.0040	0.0080	0.0042	0.0084
Z includes a continuity correction of 0.5.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
7.0267	1	0.0080

2. People are considered overweight if their BMI is 25 or greater. Create an indicator variable that identifies whether a person is overweight or not at the end of the study. When you do this, make sure you correctly handle missing values. Does the physical activity intervention change a person's chances of being overweight? Use a chi-square test of independence to answer this question. Clarify what your null hypothesis was, what conclusion you reached (in lay, not statistical, terms), and how you reached that conclusion (test statistic/p-value). (3 points)

```
data new; set hw4;
      if endlineBMI=. then overweight=.;
            else if endlineBMI>=25 then overweight=1;
            else overweight=0;
run;
proc freq data=new;
      tables treatment*overweight / nopercent nocol chisq fisher;
run;
```

Chi-square test

H_0: the proportion of people who are overweight is the same whether or not people received the intervention

p=0.0979 (greater than α)

This study does not provide evidence at the 0.05 significance level that the proportion of people who are overweight differs depending on whether or not they received the intervention

3. Answer #2 using a <u>nonparametric</u> test. Clarify which test you used, what your null hypothesis was, what conclusion you reached (in lay, not statistical, terms), and how you reached that conclusion (test statistic/p-value). (4 points)

```
proc freq data=new;
      tables treatment*overweight / nopercent nocol chisq fisher;
run;
```

Fisher's exact test
H_0: the proportion of people who are overweight is the same whether or not people received the intervention
p=0.1062 (greater than α)
This study does not provide evidence at the 0.05 significance level that the proportion of people who are overweight differs depending on whether or not they received the intervention

4. Does the BMI of people who receive no intervention change over time? Use a parametric test to answer this question. Clarify which test you used, what your null hypothesis was, what conclusion you reached (in lay, not statistical, terms), and how you reached that conclusion (test statistic/p-value). (4 points)

Use a paired t-test on the control group data (treatment=0)
H_0: there is no change in average BMI over time among people who receive no intervention
p<0.0001 (less than α)
This study provides sufficient evidence at the 0.05 significance level that there is a change in average BMI over time among people who receive no intervention

```
data new; set new;
      diff = endlineBMI-baselineBMI;
run;

proc ttest plots(showh0) data=new;
      where treatment=0;
      var diff;
run;
```

5. Does the BMI of people who receive the physical activity intervention change over time? Use a nonparametric test to answer this question. Clarify which test you used, what your null hypothesis was, what conclusion you reached (in lay, not statistical, terms), and how you reached that conclusion (test statistic/p-value). (4 points)

Use a Wilcoxon signed-rank test on the treatment group data (treatment=1)
H_0: there is no change in average BMI over time among people who receive the intervention
P<0.0001 (less than α)
This study provides sufficient evidence at the 0.05 significance level that there is a change in average BMI over time among people who receive the intervention

```
proc univariate data=new;
      where treatment=1;
      var diff;
run;
```

6. You want to evaluate a new integrated diet and physical activity intervention in a similar population using a randomized trial with a completely randomized design. This intervention will be compared to a no-intervention control. You want to detect a difference in BMI of at least 1 kg/m^2 between the two treatments. What sample size would you need to have at least 80% power? 90% power? Use the BMI data at baseline to help you conduct the power calculations. Assume a two-sided hypothesis and a significance level of 0.05. (3 points)

The standard deviation of baseline BMI is 1.7933285. We can use this for our power calculations:

```
proc power;
      twosamplemeans
            meandiff     = 1
            stddev       = 1.79
            power        = 0.8 0.9
            alpha        = 0.05
            sides        = 2
            nulldiff     = 0
            npergroup    = .;
      ods output output=output_meandiffs;
run;
```

For 80% power, we need 52 participants per group. For 90% power, we need 69 participants per group. Consider increasing these numbers to allow 10% loss to follow-up.

7. [Note this question is not related to the others above.] You have designed an intervention that aims to reduce premature birth in a vulnerable population. You will conduct a randomized trial in which you will compare pregnancy outcomes of women receiving your intervention to those of women receiving standard antenatal care. Your outcome of interest is gestational age at delivery, measured in weeks. You conduct a power calculation assuming a 0.05 significance level and determine you need 253 women in each study arm.

a) If you increase the number of women in each study arm, will your statistical power increase or decrease? (1 point)

Power will increase with sample size (largely because you will reduce your standard errors).

b) You decide you want to detect a smaller effect size than you used in your original power calculations. You will still use the same sample size. Will your statistical power now increase or decrease? (1 point)

Power will decrease (if you're looking for something smaller, you're less likely to find it).

c) You decide that you would rather use a 0.01 significance level. Will your statistical power now increase or decrease? (1 point)

Power will decrease (you're raising the bar for rejecting H_0).